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Equity Research and AI-Augmented Investment Decision Framework for the Indian Insurance Sector

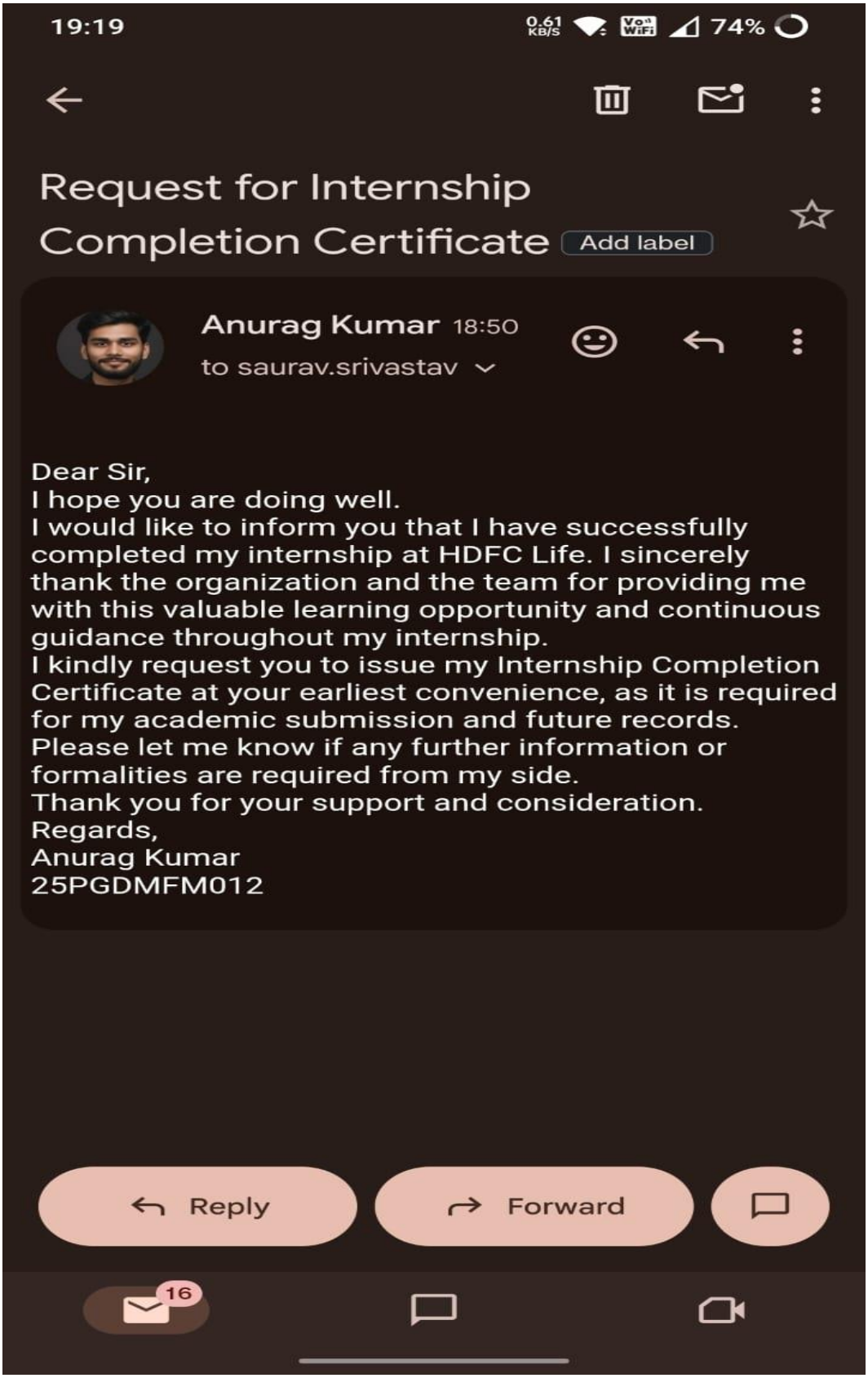
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CERTIFICATE OF COMPLETION

(Corporate Mentor / Organisation)



CERTIFICATE OF COMPLETION

(Faculty Mentor)

This is to certify that Mr. Anurag Kumar (Roll No. 25PGDMFM012) of PGDM Batch 2025–26 has completed the Corporate Internship Program at HDFC Life Insurance Co. Ltd. for a duration of 14.04.2026 to 04.07.2026.

Project title: Equity Research and AI-Augmented Investment Decision Framework for the Indian Insurance Sector

The report has been prepared under my guidance in fulfilment of the requirements of the PGDM programme at FIIB, New Delhi, and is in line with academic expectations relating to structure, analysis, ethical responsibility, and professional presentation.

The student has also completed the AI deliverable referenced in the AI Reference Section of this report, submitted separately on FIIB HUB as required.

Faculty Mentor name: Dr. Sudhi Sharma Designation: _____

Signature _____ Date _____

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I am grateful to HDFC Life Insurance Co. Ltd. for the opportunity to spend my summer internship inside a business I had, until then, only studied from the outside. Working within the Equity Research and Wealth Management function while simultaneously learning to evaluate insurers and banks as an investor gave me a vantage point that shaped almost every argument in this report.

My thanks go first to my Corporate Mentor, Mr. Saurav Srivastav, for the time he gave to my questions, for reviewing the AI-assisted recommendation tool described in the AI Solution Case Write-Up, and for the candid feedback on the divergences it produced. I am equally grateful to my Faculty Mentor, Dr. Sudhi Sharma, whose guidance sharpened the research question and the structure of this report.

I would also like to thank the colleagues at HDFC Life who explained the mechanics of insurance research patiently, and my faculty at FIIB, whose courses in financial statement analysis, valuation, financial markets, and applied technology gave me the toolkit this project depended on.

Anurag Kumar
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Executive Summary

This report presents an equity research and investment decision framework developed during a Corporate Internship Programme at HDFC Life Insurance Co. Ltd. The study focuses on the Indian insurance sector, covering both life and general insurance companies listed on the NSE, and is supported by an AI-assisted equity research automation tool built and tested during the internship.

Five key financial ratios Return on Equity, Combined Ratio, Persistency Ratio, Solvency Ratio, and Embedded Value Growth were used to assess the fundamental quality of five NSE-listed insurers spanning life and general insurance, private and public ownership: HDFC Life, SBI Life, ICICI Prudential Life, LIC of India, and GIC Re. A ten-indicator technical framework was applied and empirically back-tested, establishing that only the Moving Average and, to a lesser extent, Close Price action carry positive predictive reliability for this sector, while several widely used indicators, including RSI and VWAP, do not clear a 50% win probability as standalone signals.

The fundamental and technical dimensions were combined into an integrated stock-selection framework requiring both financial strength and favourable technical positioning. Because traditional DCF models misprice insurers, given the difficulty of separating operating and financing activity and the multi-year emergence of policy value, this study instead applies Embedded Value, Price-to-Embedded Value, and Price-to-Book approaches. LIC of India and GIC Re emerged as the strongest candidates across all three analytical dimensions and form the core holdings of the recommended portfolio, with HDFC Life, SBI Life, and ICICI Prudential Life retained on a watch list as high-quality businesses awaiting more attractive entry points.

The report also documents six screening frameworks implemented as live Smallcase portfolios, and as the internship's formal AI deliverable a working prototype AI-assisted equity research tool built on the Google Gemini API. The tool automates Buy/Hold/Avoid recommendations across eleven NSE-listed banking stocks by applying a consistent, back-tested indicator-weighting framework, and was tested end-to-end and demonstrated to the corporate mentor, achieving 72.7% alignment with manually derived analyst views and surfacing analytically valuable divergences on the remaining names. Together, these elements reflect the breadth of learning during the internship spanning financial analysis, portfolio construction, and applied AI in equity research.

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Abbreviations

Abbreviation	Expansion
ADX	Average Directional Index
API	Application Programming Interface
CAR	Capital Adequacy Ratio
DCF	Discounted Cash Flow
EV	Embedded Value
FCFF	Free Cash Flow to the Firm
GIC	General Insurance Corporation of India
IRDAI	Insurance Regulatory and Development Authority of India
KYC	Know Your Customer
LIC	Life Insurance Corporation of India
MA	Moving Average
MACD	Moving Average Convergence Divergence
NIM	Net Interest Margin
NPA	Non-Performing Asset
P/B	Price to Book
P/EV	Price to Embedded Value
PEG	Price/Earnings-to-Growth ratio
ROA	Return on Assets
ROE	Return on Equity
RSI	Relative Strength Index
VNB	Value of New Business
VWAP	Volume Weighted Average Price
WACC	Weighted Average Cost of Capital

Chapter 1: Introduction and Industry Context

1.1 Why This Study and Why Now

The Indian insurance sector is at an inflection point. Regulatory reforms under IRDAI, rising awareness of life and health coverage following the COVID-19 pandemic, and the expansion of digital distribution channels have collectively accelerated growth across the sector. Yet despite this momentum, the investment community often evaluates insurance stocks using frameworks borrowed from non-financial industries, which can be misleading and can obscure the true value embedded in insurance businesses.

At the same time, the proliferation of technical indicators in equity research has created a different problem: analysts often apply multiple signals without any empirical basis for choosing between them. This study addresses both gaps by building a sector-appropriate fundamental framework and by establishing an evidence-based hierarchy of technical signals grounded in back-tested win probabilities.

The timing of this study is also relevant. India's insurance penetration remains low compared to global peers, suggesting long-term structural growth potential. Against this backdrop, identifying which companies combine financial strength with attractive valuation represents a meaningful and practically relevant research question.

1.2 The Host Organisation and My Role

This project was carried out during my internship with HDFC Life Insurance Co. Ltd., one of India's leading private life insurance companies. I worked as an Equity Research and Wealth Management Intern, where I was exposed to various aspects of investment research, portfolio analysis, valuation, and market-based decision-making.

The internship provided a unique perspective. While working within an insurance company, I was also evaluating insurance companies from the viewpoint of an investor. This helped me understand how insurers create value through underwriting, policy persistency, risk management, and investment activities, while also assessing whether these strengths are reflected in their market performance and valuation.

It is important to note that this report does not use any confidential or internal information from HDFC Life. All analysis is based solely on publicly available data, including annual reports, financial statements, stock market information, and regulatory disclosures. This approach ensures that the conclusions and recommendations can be independently verified by any investor.

1.3 The Problem This Report Seeks to Address

Equity research on Indian insurance stocks faces two structural challenges. First, the financial metrics commonly applied to non-financial companies such as earnings per share, free cash flow, and standard P/E ratios do not translate cleanly to insurance businesses. Insurance companies generate value through long-duration policyholder relationships, investment income, and actuarially managed liabilities. A valuation model that ignores these dynamics will consistently misprice insurers.

Second, investors and analysts who do apply technical analysis to insurance stocks typically treat all indicators as equally reliable. There is rarely any empirical testing of which signals have historically performed best in this sector. As a result, investment decisions may be influenced by technically weak or even negatively predictive signals.

This report seeks to address both problems. It builds a sector-appropriate fundamental screening framework, identifies which technical indicators carry genuine predictive value for insurance stocks through back-testing, and combines the two into an integrated investment decision framework. It also extends the same evidence-based logic into an AI-assisted research tool, described in the AI Reference Section and Annexure D, which applies a back-tested indicator hierarchy consistently across a wider coverage universe of banking stocks.

1.4 Companies Included in the Study

The study covers five insurance companies listed on the National Stock Exchange of India. These include representatives from both the life and general insurance segments, spanning private and public sector ownership. The selected companies are:

- HDFC Life Insurance
- SBI Life Insurance
- ICICI Prudential Life Insurance
- Life Insurance Corporation of India (LIC)
- General Insurance Corporation of India (GIC Re)

This selection provides meaningful diversity across business models, ownership structures, and financial profiles, enabling a comparative analysis that is representative of the broader insurance sector.

1.5 Organisation of the Report

The report proceeds through eleven substantive chapters. Chapter 2 defines the objectives, scope, and research methodology. Chapter 3 presents the fundamental analysis, including ratio selection, scoring, and company-level assessment. Chapter 4 describes the technical back-testing exercise and its results. Chapter 5 combines fundamental and technical findings to arrive at stock selections. Chapter 6 addresses valuation, explaining why traditional DCF models are not appropriate for insurers and describing the approaches used instead. Chapter 7 presents the final portfolio recommendation. Chapter 8 documents the six screening frameworks and their Smallcase implementation. Chapter 9 proposes an AI-assisted underwriting triage workflow. Chapter 10 presents conclusions and recommendations, and Chapter 11 reflects on the professional learning from the internship. The AI Reference Section, References, and Annexures follow, with the full AI Solution Case Write-Up and Proof Bundle for the AI-assisted equity research tool at Annexure D.

Chapter 2: Objectives, Scope, and Methodology

2.1 Objectives

This study has four primary objectives:

- To evaluate the fundamental financial quality of selected NSE-listed insurance companies using a structured ratio-based framework.
- To assess the reliability of technical indicators for insurance stocks through empirical back-testing.
- To combine fundamental quality and technical evidence into an integrated stock selection process.
- To apply sector-appropriate valuation methods to determine whether the selected companies offer attractive long-term investment potential.

A secondary objective involves the development of six screening frameworks and their implementation as live Smallcase portfolios, along with the design of an AI-assisted underwriting workflow and the build of a working AI-assisted equity research tool as the internship's formal AI deliverable.

2.2 Scope and Boundaries

The study covers five insurance companies for the period FY22–FY24. Only publicly available data sources were used. The analysis is focused on equity investment decision-making and does not extend to fixed income, derivatives, or alternative investment instruments. Valuation conclusions are based on relative methods rather than absolute intrinsic value estimates, given the structural limitations of DCF for insurance businesses.

2.3 Research Framework

The analytical framework proceeds in three integrated stages:

Stage	Tool used	Output
Quality	Five-ratio fundamental scorecard (ROE, Combined Ratio, Persistency, Solvency, EV Growth)	Comparative assessment of the five insurers
Timing	Back-testing of ten technical indicators	Evidence-based indicator reliability hierarchy
Decision	Integrated fundamental + technical + valuation screen	Core holdings and watch list

Table 2.1 Research framework: three integrated stages.

2.4 Data Sources

Data was sourced from company annual reports, NSE and BSE regulatory filings, IRDAI publications, Screener.in, TradingView, and publicly available investor presentations. All data used in this report is in the public domain and independently verifiable.

2.5 Limitations of the Study

The study covers a three-year data window (FY22–FY24), which may not capture complete business cycles. The back-testing exercise, while empirically grounded, is based on historical patterns and does not guarantee future signal performance. The Smallcase tracking period is short a few weeks and early returns are not suitable for drawing conclusions about long-term strategy performance. Valuation estimates are based on publicly reported embedded value figures and relative multiples, and do not involve independent actuarial assessment.

Chapter 3: Fundamental Analysis of the Insurance Sector

3.1 Why These Five Ratios

Insurance companies require a tailored set of financial metrics. Five ratios were selected for this study on the basis of their relevance to insurance business models and their ability to capture different dimensions of performance:

Ratio	What it captures
Return on Equity (ROE)	Profitability how hard shareholders' capital is working
Combined Ratio (general insurers)	Underwriting efficiency and cost discipline
Persistency Ratio (life insurers)	Policy retention and book quality
Solvency Ratio	Financial resilience against the regulatory minimum
Embedded Value Growth	Value creation from the in-force and new business

Table 3.1 The five insurance ratios and their rationale.

Together, these metrics assess profitability, underwriting efficiency, policy retention, financial resilience, and value creation. No single ratio is sufficient on its own; the combination provides a more complete picture of business quality.

3.2 Reading the Numbers

Each ratio was computed for the five companies over FY22–FY24 using data from annual reports and regulatory disclosures. Where a ratio is not directly applicable to one segment for example, the combined ratio is primarily relevant for general insurers a suitable substitute was used, and the methodology is noted accordingly.

3.3 The Ranking Method, and a Correction Worth Explaining

Companies were ranked on each ratio, with the highest-performing company in each metric receiving the best score. A composite score was then computed by averaging the individual rankings. One correction is worth noting: for the solvency ratio, a very high figure is not always an indicator of superior performance. An excessively high solvency ratio may suggest that capital is not being deployed efficiently. Companies were therefore assessed on whether their solvency ratio was comfortably above the regulatory minimum rather than on absolute magnitude alone.

3.4 What the Fundamentals Say

The fundamental analysis reveals clear differentiation across the five companies. LIC of India stands out for its scale, persistency, and embedded value, while GIC Re demonstrates strong underwriting discipline and solvency. The private life insurers HDFC Life, SBI Life, and ICICI Prudential show competitive embedded value growth and persistency ratios, reflecting the improving quality of their business mix. Across all five companies, profitability and capital adequacy remain sound, though the private insurers trade at premium valuations relative to their public-sector counterparts.

3.5 Reading Life and General Insurers on Their Own Terms

Life and general insurance businesses are structurally different and should not be evaluated using the same framework. Life insurers derive long-term value from policy persistency, investment income on reserves, and the embedded profitability of the in-force book. General insurers are evaluated primarily on underwriting profitability (combined ratio) and investment returns. This report applies distinct analytical lenses to each segment and avoids direct cross-segment comparisons where the underlying economics differ significantly.

3.6 Company-by-Company Snapshot

HDFC Life Insurance

Demonstrates strong new business margins, high persistency, and consistent embedded value growth.

SBI Life Insurance

Benefits from its bancassurance network and has shown steady improvement in both VNB and persistency.

ICICI Prudential Life Insurance

Has focused on improving its product mix toward protection and non-par savings, which has supported margin expansion.

LIC of India

Combines unmatched scale with improving solvency and persistency ratios, and its embedded value represents a very large pool of future economic value.

GIC Re

Maintains a combined ratio that is broadly in line with global reinsurance peers and demonstrates prudent capital management.

Chapter 4: Technical Analysis and Back-testing

4.1 What Was Tested and How

A ten-indicator framework was applied to the selected insurance stocks using historical price data sourced from TradingView and NSE archives. The indicators tested were: Moving Average (MA), Close Price action (CP), Supertrend, RSI, VWAP, MACD, Bollinger Bands, Stochastic Oscillator, ADX, and volume-based signals. For each indicator, signals were generated historically and the win probability the proportion of times a signal led to a correct directional outcome was computed.

4.2 Results of the Back-testing Exercise

The back-testing results reveal a clear hierarchy of indicator reliability:

Indicator	Win probability	Role assigned
Moving Average	68.43%	Primary signal
Close Price action	53.25%	Secondary confirmation
Supertrend	49.15%	Supplementary
RSI	46.51%	Supplementary only not primary
VWAP	44.18%	Least reliable use with caution
MACD, Bollinger Bands, Stochastic, ADX, Volume	Below 50%	Not used as standalone triggers

Table 4.1 Back-tested win probability of ten technical indicators.

These findings are consistent with back-testing results observed in the banking sector analysis undertaken during the same internship (see Annexure D), suggesting that this indicator hierarchy may have broader applicability across financial sector stocks.

4.3 Implications of the Findings

The most practically important implication is that widely used indicators such as RSI and VWAP should not be used as primary triggers for investment decisions in insurance stocks. Their sub-50% win probabilities mean that acting on them in isolation is likely to produce worse outcomes than not using them at all. This finding challenges the conventional practice of applying all indicators with equal weight. The evidence-based hierarchy established through back-testing MA first, CP second, Supertrend supplementary, others contextual represents a more disciplined and defensible approach to technical signal integration. This same hierarchy was carried forward directly into the prompt design of the AI-assisted equity research tool described in Annexure D.

4.4 Quarter-wise Observations and Future Improvements

A quarter-wise breakdown of signal performance reveals that Moving Average signals are most reliable in trending markets and less reliable during sideways or range-bound periods. Future improvements to this framework could include conditioning indicator weights on market regime (trending versus mean-reverting) and extending the back-test window beyond the current dataset to capture a fuller range of market conditions. The framework as it currently stands is a meaningful improvement over unweighted signal application, but it is a starting point rather than a final product.

Chapter 5: Integrated Stock Selection

5.1 Bringing Quality and Value Together

The integrated stock selection process combines the fundamental scores from Chapter 3 with the technically timed signals from Chapter 4. A company qualifies for consideration only if it demonstrates financial strength on the fundamental scorecard and offers a favourable technical setup. This two-stage filter prevents the selection of fundamentally weak companies based on positive chart momentum, and avoids entering fundamentally strong companies at technically overbought levels.

5.2 Significance of the Combined Approach

The combined approach is significant because fundamental quality and technical attractiveness are independent dimensions. A company can be fundamentally excellent but technically overextended, meaning that even a correct fundamental assessment can lead to poor short-term returns if entry timing is ignored. Conversely, a company with improving technical momentum may not have the underlying financial strength to sustain its price appreciation. The integrated framework navigates this by requiring both conditions to be met simultaneously.

5.3 Investment Rationale for the Selected Companies

The integrated analysis points to LIC of India and GIC Re as the two companies that best satisfy both the fundamental quality and technical value conditions. LIC offers unmatched scale, improving persistency, strong embedded value, and a valuation that remains attractive relative to its financial profile. GIC Re combines disciplined underwriting with a solvency position that is well above regulatory requirements, and its valuation is modest relative to its profitability and capital strength. Both companies offer the kind of alignment between business quality and market price that supports a well-grounded long-term investment case.

Chapter 6: Valuation DCF and Why Insurers Are Different

6.1 Understanding the DCF Approach

One of the important learning experiences during the internship was understanding how to build a Discounted Cash Flow (DCF) model from scratch. As part of the training process, a complete three-statement financial model and DCF valuation was built using Alembic Pharmaceuticals. A non-financial company was chosen because it provides a straightforward environment for learning the fundamentals of valuation.

The key steps involved in a DCF model include: forecasting financial performance, estimating Free Cash Flow to the Firm (FCFF), discounting future cash flows using the Weighted Average Cost of Capital (WACC), calculating terminal value, and ultimately arriving at an estimate of intrinsic value (Damodaran, 2012).

Although the exercise helped in understanding the mechanics of valuation, it also highlighted an important point: the traditional DCF framework that works well for manufacturing and service companies is not always suitable for insurance companies. This realization became an important part of the learning process and shaped the valuation approach used in this study.

6.2 Why a Traditional DCF Has Limitations for Insurers

A conventional FCFF-based DCF model assumes that operating activities and financing activities can be separated clearly. For most non-financial companies, this distinction is relatively straightforward. However, insurance companies operate differently.

For an insurer, collecting premiums, managing policy liabilities, and investing funds are all core business activities. As a result, separating operating cash flows from financing cash flows becomes difficult and can lead to valuation estimates that do not fully reflect the economics of the business.

Another challenge arises particularly in the case of life insurance companies. The value created when a policy is sold is often realised gradually over many years. Therefore, the profits reported in a single financial year may not fully capture the economic value of the business. A valuation model based only on current-period cash flows may overlook a significant portion of the value embedded within the existing insurance portfolio.

Because of these characteristics, insurance companies are generally valued using approaches that are specifically designed for the sector rather than relying solely on traditional FCFF-based DCF models.

6.3 Valuation Approaches Used for Insurance Companies

Segment	Approach used	Why
Life insurers	Embedded Value (EV) and Value of New Business (VNB)	Captures net worth plus present value of future profits from existing and new business
General insurers	Dividend discount, excess return, and relative valuation models	Underwriting-led economics do not fit FCFF-style modelling
Both segments	Price-to-Embedded Value (P/EV) and Price-to-Book (P/BV)	Sector-appropriate multiples used widely by market participants

Table 6.1 Valuation approaches used for life and general insurers.

For the two companies selected in this study, LIC of India and GIC of India, valuation was therefore assessed using a relative valuation approach rather than applying a traditional DCF model. LIC appears attractively valued when viewed against its strong return on equity and growth profile. Similarly, GIC combines strong solvency, profitability, and investment performance with a valuation that remains reasonable relative to its fundamentals (Damodaran, 2012).

6.4 Investment Decision Framework

The final investment framework used in this study combines three elements: business quality, valuation attractiveness, and sector-appropriate valuation methods.

A low valuation multiple by itself is not sufficient to justify an investment if the underlying business lacks financial strength. Similarly, a high-quality company may not always be an attractive investment if its future growth expectations are already fully reflected in its market price.

The objective, therefore, is to identify companies where business quality and valuation are aligned. Based on the analysis conducted in this report, GIC of India and LIC of India satisfied these conditions more effectively than the other insurers included in the study. These two companies form the foundation of the recommended portfolio presented in the next chapter.

Chapter 7: Portfolio Recommendation

The recommended portfolio concentrates on LIC of India and GIC Re as the core holdings, with the remaining three companies HDFC Life, SBI Life, and ICICI Prudential acknowledged as high-quality businesses that may offer entry opportunities at more attractive valuations.

The portfolio reflects the full analytical process: companies were not selected on any single criterion but on the convergence of fundamental quality, technical timing, and valuation reasonableness. This multi-lens approach reduces the risk of confirmation bias and ensures that each holding can be justified on more than one independent dimension.

Category	Company	Rationale
Core holding	LIC of India	Unmatched scale, strong embedded value, improving persistency, attractive relative valuation
Core holding	GIC Re	Disciplined underwriting, robust solvency, modest valuation relative to profitability
Watch list	HDFC Life Insurance	Strong new business margins and persistency; await a more attractive entry point
Watch list	SBI Life Insurance	Bancassurance strength and steady VNB improvement; currently fully valued
Watch list	ICICI Prudential Life	Improving product mix toward protection; currently fully valued

Table 7.1 Recommended portfolio: core holdings and watch list.

Chapter 8: Screening Frameworks and Smallcase Implementation

8.1 Why Screening, and the Different Investment Approaches

While the earlier chapters focused on analysing insurance companies, another important part of the internship involved learning how investors identify opportunities across a much larger universe of stocks. A stock screener allows investors to apply predefined financial criteria to a large number of companies and identify those that match a particular investment philosophy.

Six different screening frameworks were developed based on well-known investment approaches. Each framework was then implemented as a Smallcase portfolio so that its performance could be tracked over time.

- Institutional Growth large, established companies with consistent growth and characteristics preferred by institutional investors.
- Buffett Value inspired by Warren Buffett's principles; financially strong businesses with sustainable competitive advantages and reasonable valuations.
- Peter Lynch GARP Growth at a Reasonable Price; emphasises the PEG ratio to find companies growing steadily without excessive valuation.
- Growth Investing targets companies with strong revenue and earnings growth, even at relatively higher valuation multiples.
- Multibagger Discovery focuses on smaller companies with strong growth potential and higher acceptable risk levels.
- High ROCE and Low PE identifies companies generating efficient returns on capital while trading at comparatively lower valuation multiples.

8.2 Comparative Evaluation and Smallcase Tracking

After creating the screening frameworks, each one was implemented and tracked through Smallcase portfolios. Because the portfolios were created in May 2026, the available data covers only a few weeks. Results should therefore be viewed as preliminary observations rather than evidence of long-term strategy performance.

Framework	Return since inception	Since
Institutional Growth (Method-1)	+0.35%	20 May 2026
Buffett Value (Method-2)	−1.01%	23 May 2026
Peter Lynch GARP (Method-3)	−2.64%	26 May 2026
Growth Stock	−1.10%	22 May 2026
Multibagger Discovery	+2.62%	23 May 2026
High ROCE and Low PE (Method-6)	−3.65%	25 May 2026

Note: Observation windows of only a few weeks; returns are indicative only and are not comparable across frameworks due to different start dates.

8.3 Key Learnings from the Screening Exercise

At first glance, the Multibagger Discovery framework appears to have delivered the strongest return during the observation period. However, drawing conclusions based solely on a few weeks of performance would not be appropriate. A meaningful comparison between investment strategies requires a longer evaluation period, ideally across different market conditions and economic cycles.

From a theoretical perspective, quality- and value-based approaches such as Buffett Value, Peter Lynch GARP, and High ROCE with Low PE continue to have strong support in both academic research and investment practice. Although their short-term returns may fluctuate, these strategies are often associated with long-term wealth creation.

The most valuable outcome of this exercise was not the early performance numbers but the process itself building the screening frameworks helped translate different investment philosophies into measurable financial criteria and provided practical exposure to portfolio monitoring and performance evaluation.

Chapter 9: A Proposed AI-Assisted Underwriting Triage Workflow

9.1 Background of the Idea

During the internship, aspects of HDFC Life's operations beyond equity research were observed. One area that stood out as a candidate for process improvement was term insurance underwriting. The current underwriting process is document-intensive and largely manual, which creates delays in policy issuance and introduces variability in risk assessment outcomes. It should be stated clearly that what follows is a proposed model, developed for academic purposes, and does not represent any process used by HDFC Life Insurance Co. Ltd.

9.2 Understanding the Problem

Term insurance underwriting requires the collection and cross-validation of multiple documents, including KYC forms, income proof, bank statements, medical examination reports, and income tax returns. Each of these documents must be reviewed individually and checked for consistency. In high-volume periods, manual processing creates bottlenecks, and the risk of human error increases. Turnaround time from application to policy issuance is consequently longer than it could be with automated support.

9.3 Proposed AI-Assisted Workflow

The proposed workflow uses the following modules in sequence:

1. OCR extraction: structured data extracted from documents at the point of submission.
2. Cross-document validation: algorithms check for consistency across KYC information, income declarations, and bank statement data.
3. Income eligibility module: assesses whether the sum assured requested is within appropriate limits given verified income.
4. Medical risk assessment module: flags applications where reported health details require closer review.
5. Risk-scoring engine: categorises each application as (a) straightforward approval, (b) further review required, or (c) escalation to a senior underwriter.

9.4 Demonstration, Benefits, and Limitations

The primary benefit of this workflow is a reduction in processing time for straightforward applications, which represent the majority of term insurance submissions. Human underwriters are freed from routine document validation and can focus their expertise on genuinely complex cases. The proposal also reduces the variability that arises when different reviewers apply different thresholds to similar cases.

The main limitation is that AI-based document processing is only as reliable as the quality of the input data. Poor-quality scans, handwritten forms, or documents in non-standard formats may reduce extraction accuracy. The system would also require periodic retraining as document formats and underwriting guidelines evolve.

9.5 Current Status and Future Scope

The proposal remains at the design and concept stage. No implementation has been undertaken, and no confidential HDFC Life data was used in developing the concept. Future scope could include integration with the IRDAI-mandated insurance repository, real-time medical data APIs, and bureau-based income verification services. If implemented, the workflow has the potential to meaningfully improve the customer experience for term insurance applicants while maintaining underwriting rigour.

This underwriting concept remains a design exercise rather than a working prototype. The internship's working, tested AI deliverable an AI-assisted equity research recommendation tool applying the indicator hierarchy from Chapter 4 across a wider banking-stock universe is documented in full in the AI Reference Section below and in Annexure D.

Chapter 10: Conclusions and Recommendations

10.1 Principal Conclusions

The analysis presented in this report leads to three principal conclusions.

First, fundamental analysis of insurance companies must use sector-specific metrics. Standard corporate finance ratios, applied without adaptation, will produce misleading assessments of insurance business quality. The five-ratio framework used here covering profitability, persistency, underwriting quality, solvency, and embedded value growth provides a more reliable basis for comparison.

Second, technical indicators are not equally reliable. The back-testing evidence establishes that the Moving Average is the strongest signal for insurance stocks, while RSI and VWAP have negative expected value when used as standalone triggers. Investors and analysts should adopt an evidence-weighted approach to technical signals rather than treating them symmetrically a principle that was carried forward directly into the design of the AI-assisted recommendation tool documented in Annexure D.

Third, LIC of India and GIC Re represent the most compelling investment opportunities within the current insurance sector universe. Both companies combine financial strength with valuation that has not fully reflected their fundamental quality. The integrated framework rather than any single analytical dimension supports this conclusion.

10.2 Recommendations

- Use Embedded Value and VNB growth as the primary lens for assessing life insurers, supplemented by persistency ratios as a measure of business quality durability.
- For general insurers, anchor valuation on the combined ratio and solvency position.
- When applying technical analysis, weight Moving Average signals most heavily and treat RSI and VWAP signals as contextual rather than actionable on their own.
- Enter positions where both fundamental quality and technical timing support the case.
- LIC and GIC Re are the recommended initial holdings based on the analysis in this report.
- Extend the AI-assisted recommendation framework documented in Annexure D from banking stocks to the insurance coverage universe in this report, so that fundamental, technical, and AI-generated views can be compared directly for the same companies.

Chapter 11: Professional Reflection and Employability

11.1 Key Learnings from the Internship

The internship required working across three distinct analytical disciplines fundamental financial analysis, technical chart-reading, and empirical back-testing and integrating them into a single coherent framework. This required developing comfort with ambiguity and with the discipline of testing assumptions rather than accepting them. The finding that widely used technical indicators like RSI have below-50% win probabilities was a genuinely counterintuitive result, and working through its implications reinforced the value of evidence-based reasoning over convention.

The exposure to insurance-specific financial concepts embedded value, value of new business, persistency, solvency margins was practically valuable and directly applicable to research roles in the BFSI sector. Building and testing the AI-assisted recommendation tool described in Annexure D added a further, complementary skill: translating a back-tested analytical framework into a structured, repeatable AI workflow, and interpreting the cases where the AI and a human analyst genuinely disagreed.

11.2 Application of Academic Concepts

The internship provided an opportunity to apply several concepts studied in the academic programme. Discounted cash flow valuation, portfolio theory, financial statement analysis, and risk management principles all featured in the work undertaken. The experience of adapting these frameworks to the specific characteristics of insurance companies rather than applying them mechanically deepened understanding of both the concepts and their practical boundaries.

11.3 Employability and Professional Readiness

The skills developed during this internship are directly relevant to research analyst, investment analyst, and strategy associate roles in financial services. The ability to build structured analytical frameworks, test assumptions empirically, and communicate findings clearly and concisely are transferable across a range of functions within BFSI and beyond. The habit of questioning whether a commonly used tool actually performs as expected and testing it rather than assuming is a professional discipline that will apply in any analytical role. Exposure to applied AI, from prompt design to comparing AI output against expert judgement, is an increasingly relevant addition to that skill set.

AI Reference Section

In accordance with the Corporate Internship Program requirements, the AI deliverable associated with this internship is submitted separately on FIIB HUB and is referenced here. The full case write-up and proof bundle are reproduced at Annexure D.

AI deliverable title	AI-Assisted Equity Research Recommendation Tool (Gemini-Powered, Indicator-Weighted)
Route	AI Solution Case Write-Up and Proof Bundle
Problem addressed	Repetitive, individually-assessed fundamental-plus-technical review of eleven NSE-listed banking stocks, with inconsistency risk across analysts and iterations
Solution built	A prompt-engineering pipeline on the Google Gemini API that ingests seven fundamental ratios and one technical signal per bank and returns a structured Buy / Hold / Avoid recommendation, a 150–200 word reasoning note, and a signal-tension note, using the back-tested indicator-weighting hierarchy from Chapter 4
Working prototype	Python script (ai_research.py) reading bank_data.xlsx and writing AI_Recommendations_Output.xlsx
Data used	Publicly sourced fundamental ratios and technical signals for 11 NSE-listed banks (Screener.in, NSE/BSE filings, TradingView, annual reports FY22–24). No confidential HDFC Life data was used
Deployment level claimed	Level 1 demonstrated in context to the Corporate Mentor, Mr. Saurav Srivastav, in Week 8
Proof submitted	Working script, sample input/output, full 11-bank testing summary, and supervisor feedback on divergent cases (Annexure D)
Date of submission	July 2026
Where discussed in this report	Chapter 4 (indicator hierarchy); full write-up and proof at Annexure D

Reference

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Annexure A- Compliance and Similarity Declaration

I declare the following in respect of this report.

- The analysis rests entirely on publicly available information company annual reports, financial statements, exchange filings, market data, and regulatory disclosures. No confidential or proprietary information belonging to HDFC Life Insurance Co. Ltd. has been used at any point.
- The AI-assisted equity research tool described in the AI Reference Section and Annexure D was built and tested using publicly sourced fundamental and technical data for eleven NSE-listed banks. No confidential HDFC Life data was accessed or processed.
- The AI-assisted underwriting triage concept in Chapter 9 is a design proposal only. No implementation has been undertaken, and no confidential data was used in developing it.
- The portfolio presented in Chapter 7 is illustrative and is not a live, funded investment recommendation.
- All references and in-text citations follow APA 7th edition.
- The plagiarism similarity report (Annexure B) and the artificial intelligence similarity report (Annexure C) are appended, and the AI Solution Case Write-Up and Proof Bundle appears at Annexure D.

Signature _____ Date _____

Anurag Kumar, Roll No. 25PGDMFM012

Annexure B - Plagiarism Similarity Report

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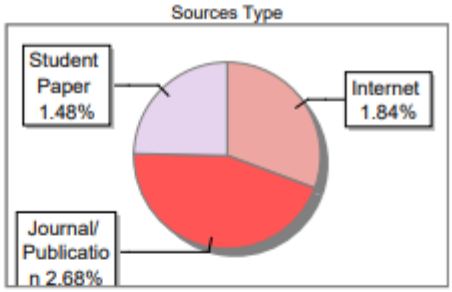
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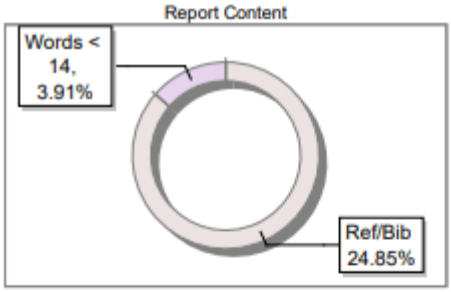
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Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
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Annexure C- Artificial Intelligence Similarity Report

DrillBit

AI Content Detection Report
Submission ID: 6318212

AI TEXT DETECTED (%)
Percentage of content showing AI-generated patterns

11.09

Matched 986 words showing AI-pattern signal

POSITION ON SCALE
Where this score falls on a 0 - 100% range

0% 100%

11.09 %

01 Submission Information

Document metadata

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Annexure D -AI Solution Case Write-Up and Proof Bundle

Basic Details

Project Title	Equity Research and AI-Augmented Investment Decision Framework for the Indian Insurance Sector
AI Case Title	AI-Assisted Equity Research Recommendation Tool (Gemini-Powered, Indicator-Weighted)
Host Organisation	HDFC Life Insurance Co. Ltd.
Role	Equity Research and Wealth Management Intern
Student Name	Anurag Kumar
Roll Number	25PGDMFM012
Corporate Mentor	Mr. Saurav Srivastav
Faculty Mentor	Dr. Sudhi Sharma
Submission Type	AI Case Write-Up and Proof Bundle
Deployment Status	Level 1 demonstrated in context to the Corporate Mentor; not yet used independently by another team member or adopted organisation-wide

Section 1 AI Problem Fitment

1.1 What is the problem? Who does it affect, and how?

Analysing multiple banking stocks across both fundamental and technical dimensions is a structurally repetitive task. Each bank must be individually assessed on Net Interest Margin (NIM), Return on Equity (ROE), Return on Assets (ROA), Gross and Net NPA, Capital Adequacy Ratio (CAR), and Price-to-Book (P/B), then reviewed through technical indicators. With eleven banks in the coverage universe, this means executing the same interpretive process eleven times from scratch, with inconsistency risk increasing across iterations. The problem affects the equity research team through analyst time and consistency of output, and affects the wealth management function downstream through the reliability of the recommendations it relies on.

1.2 How is it currently solved without AI?

The manual equity research process at HDFC Life requires analysts to individually assess each of the eleven banking stocks across both fundamental and technical dimensions. Each analyst applies judgement bank by bank, weighting ratios and technical signals informally, which is time-consuming at scale and creates a structurally repetitive process with a real risk of inconsistency between analysts and between review cycles for the same analyst.

1.3 Why is AI viable here?

The task fits AI-assisted automation well because the inputs are structured (seven numeric ratios plus one technical-signal description per bank) and the analytical logic including the indicator-weighting hierarchy established through back-testing in Chapter 4 can be encoded explicitly into a prompt. AI automation addresses the problem by applying identical analytical logic to every bank, producing comparable outputs rapidly, and surfacing divergences between the AI view and the human view as the most analytically valuable output of the exercise, rather than treating disagreement as a failure.

1.4 How the problem was confirmed as real

The repetitive, judgement-heavy nature of covering eleven banks individually was confirmed through direct experience carrying out the manual equity research process during the internship, and was discussed with the Corporate Mentor, Mr. Saurav Srivastav, ahead of building the prototype.

Section 2 AI Solution Design and Build

2.1 What did you build?

I built a working prototype of a prompt-engineering-based AI automation pipeline. Structured financial ratio data for eleven Indian banking stocks is fed into a prompt template, which instructs the Gemini model to generate consistent Buy / Hold / Avoid investment recommendations with reasoned analysis. The core logic mirrors a human equity research analyst workflow: structured inputs (seven fundamental ratios plus one technical signal per bank) are processed by the Gemini API using a consistent analytical framework, producing a structured output Recommendation, Reasoning, and Signal Tension which is then compared against a manually derived analyst view to identify alignments and divergences.

The prompt explicitly instructs the model to act as a senior equity research analyst specialising in Indian banking stocks, to accept the seven fundamental metrics and one technical signal as structured inputs, and to apply the back-tested indicator-weighting hierarchy below rather than treating all technical signals as equally reliable.

Indicator	Win rate	Usage instruction in prompt
Moving Average	68.43%	Weight most heavily primary signal
Close Price	53.25%	Secondary confirmation signal
Supertrend	49.15%	Supplementary
RSI	46.51%	Supplementary only not primary
VWAP	44.18%	Least reliable use with caution

Table AI.1 Indicator-weighting framework used in the AI recommendation prompt.

2.2 Tools and platforms used

Component	Details
Language	Python 3.14
AI API	Google Gemini API (gemini-2.0-flash)
Libraries	google-genai, pandas, openpyxl
Input source	Microsoft Excel (bank_data.xlsx)
Output	AI_Recommendations_Output.xlsx
Environment	Windows 11, VS Code / PowerShell

2.3 Data and documents used

The input file, bank_data.xlsx, contains eleven banks by eight columns: Bank, NIM, ROE, ROA, Gross NPA, Net NPA, CAR, P/B Ratio, and Technical Signal, compiled from annual reports (FY22–24), Screener.in, NSE/BSE filings, and TradingView. All data used is publicly available; no confidential HDFC Life data was accessed or processed at any stage.

2.4 Output produced

For each bank, the pipeline produces a structured recommendation (Buy / Hold / Avoid), a 150–200 word reasoning note grounded in the fundamental ratios and the weighted technical signal, and a signal-tension note flagging any conflict between fundamentals and technicals. The output is written to AI_Recommendations_Output.xlsx alongside the manually derived analyst view, an alignment flag, and a divergence note. A worked example for ICICI Bank is shown below.

Field	Value
Bank	ICICI Bank
NIM / ROE / ROA	4.5% / 17.8% / 2.1%
Gross NPA / Net NPA	2.3% / 0.5%
CAR / P/B	16.8% / 3.1x
Technical signal input	Above 200-day EMA, RSI at 64, bullish MACD crossover
AI recommendation	BUY
Signal tension	No material tension; fundamentals and technicals aligned. RSI approaching overbought noted as a near-term entry-timing risk only

Table AI.2 Sample AI-generated recommendation (ICICI Bank).

2.5 Limitations

- The pipeline depends on the free tier of the Gemini API, which imposes rate limits unsuitable for high-frequency or real-time use.
- Recommendations are only as reliable as the structured inputs; qualitative factors not captured in the eight input fields (management commentary, sector news, regulatory action) are not considered.
- The indicator-weighting hierarchy is based on a single back-test window (Chapter 4) and would need periodic recalibration as market conditions change.
- Divergent cases still require human judgement the tool is designed to surface disagreement, not resolve it automatically.

Section 3 AI Demonstration and Use Evidence

3.1 Deployment level achieved

Level 1 demonstrated in context. The prototype was run end-to-end across all eleven banks in the coverage universe and demonstrated to the Corporate Mentor, Mr. Saurav Srivastav, in Week 8 of the internship. It has not yet reached Level 2 (used independently by a colleague) or Level 3 (organisational adoption).

3.2 How the solution was demonstrated and tested

The script was executed successfully across all eleven banks, with output saved to AI_Recommendations_Output.xlsx. For each bank, the AI-generated recommendation was compared against a manually derived analyst view prepared independently, and each pair was marked as aligned or divergent, with a divergence note recorded for every disagreement.

3.3 Who saw or used the solution

Metric	Result
Total banks tested	11 of 11
Aligned recommendations	8 out of 11 (72.7%)
Divergent recommendations	3 out of 11 (27.3%)
Script execution	Successful output saved to Excel
Demonstrated to supervisor	Yes Mr. Saurav Srivastav, Week 8
Deployment level achieved	Level 1 demonstrated in context

Table AI.3 Testing summary: AI vs. manually derived analyst views (11 banks).

Section 4 Organisational Response and Constraint

4.1 Why the work stopped at Level 1

The tool was demonstrated to the Corporate Mentor once the eleven-bank test run was complete. That demonstration confirmed the tool works end-to-end and produces analytically defensible output, including on the three divergent cases, but it was not carried further within the internship window into independent use by another analyst or into any production workflow. As an intern working within the Equity Research and Wealth Management function, I had neither the mandate nor the timeframe to pursue organisational adoption, which would require review beyond a single supervisor demonstration.

4.2 What broader deployment would require

- Upgrading from the Gemini free tier to a production-grade API arrangement to remove rate-limit constraints.
- Formal review and sign-off from the equity research team on the indicator-weighting logic before it informs live recommendations.
- A documented model-governance framework, including an audit log of every AI recommendation and every analyst override.
- Extension and re-validation of the indicator-weighting hierarchy for the insurance coverage universe used elsewhere in this report, since the current back-test and prompt were built for banking stocks.
- Training for analysts on interpreting and acting on the tool's signal-tension notes.

4.3 Organisational engagement and observation

The Corporate Mentor, Mr. Saurav Srivastav, reviewed the eleven-bank test run in Week 8. On the IndusInd Bank divergence where the AI weighted the current Gross NPA level (2.9%) while the manual view weighted the improving three-year trajectory (down from 4.2%) he confirmed this as a genuine and realistic limitation of current-period ratio inputs, rather than an error in the tool's logic. The 72.7% alignment rate was treated as a reasonable baseline for a first prototype.

4.4 What I would do differently, and how I would take it forward

- Extend the testing window and coverage universe beyond eleven banks to build a larger evidence base before any adoption discussion.
- Add a simple dashboard front-end in place of the raw Excel output, so reviewers can see recommendations and signal-tension notes without opening a spreadsheet.
- Apply the same indicator-weighted prompt framework to the insurance coverage universe analysed in Chapters 3–7 of this report, so that AI-generated and manually derived views could be compared directly for the same companies this report covers.
- Build in a lightweight audit log from the outset, rather than adding one later, so every recommendation and override is traceable from day one.

Section 5 Proof of Solution

5.1 Proof items attached

#	Proof item	Format	Detail
1	Working script	Python file	ai_research.py reads bank_data.xlsx, calls the Gemini API, writes AI_Recommendations_Output.xlsx
2	Sample input/output	Worked example	ICICI Bank input fields and AI-generated recommendation (Table AI.2)
3	Full testing summary	Results table	11-bank alignment/divergence summary with divergence notes (Table AI.3, Section 3.3)

#	Proof item	Format	Detail
4	Supervisor engagement record	Written summary	Corporate Mentor review of the IndusInd Bank, AU Small Finance Bank, and Federal Bank divergences (Section 4.3)

5.2 Consistency check across problem, solution, deployment, response, and proof

Element	What the case states	Consistent?
Problem	Manual, per-bank review of eleven banks is repetitive and carries inconsistency risk (Section 1)	Yes
Solution	Indicator-weighted Gemini prompt pipeline producing structured recommendations (Section 2)	Yes
Deployment	Level 1 demonstrated to the Corporate Mentor in Week 8 (Section 3)	Yes
Response	Supervisor confirmed the IndusInd Bank divergence as a genuine, realistic limitation (Section 4)	Yes
Proof	Script, sample output, full testing table, and supervisor engagement record attached (Section 5.1)	Yes
Data honesty	Public data only throughout; no confidential HDFC Life data used or claimed	Yes

Every stage of the case points to the same claim, with no material contradiction.

Student Declaration

I confirm that the problem and solution described relate to a genuine operational task in equity research at HDFC Life, identified during my internship. The prototype was built and tested by me on publicly sourced data only; no confidential HDFC Life data was accessed at any stage. This deliverable was demonstrated to my Corporate Mentor, Mr. Saurav Srivastav, in Week 8, and I have not claimed any level of deployment beyond that demonstration. All sections of this bundle are internally consistent, and the proof items listed in Section 5.1 are attached.

Anurag Kumar (25PGDMFM012)

Student signature _____ Date: _____

Faculty Supervisor signature _____ Date: _____